

SIMPLETEXT 2026 · TASK 1.1 · CLEF, JENA

Prompting Based on Linguistic Analysis for Text Simplification

Nuria Pirvu · Rémi Cardon

HULAT, Universidad Carlos III de Madrid

HYPOTHESIS

Complexity comes from a few recurring phenomena.

1 · Rule-based analysis

spaCy tags each sentence's complexity features.



2 · Prompt-based generation

Each phenomenon gets its own targeted LLM prompt.

Seven tags grounded in plain-language research

Müller & François (2022): features flagged both in francophone plain-language guides and in writers' actual practice, verified on 102 administrative texts. Each maps to one simplification operation.

Tag	Target simplification
Subordination	Reduce to a single finite verb
Passive	Convert to active voice
Personal pronoun	Reinforce direct address (proximization)
Impersonal	Replace with a personal construction
Negation	Rephrase as a positive formulation
Long sentence	Shorten to at most 20 words
Acronym	Spell out or gloss the abbreviation

LANGUAGE COVERAGE

Full pipeline for five languages, single prompt for the rest

5 languages, tagged

EN · FR · ES · DE · PT

Full spaCy parsing, well-defined criteria. Each sentence is tagged, then routed to targeted prompts.

6 languages, fallback

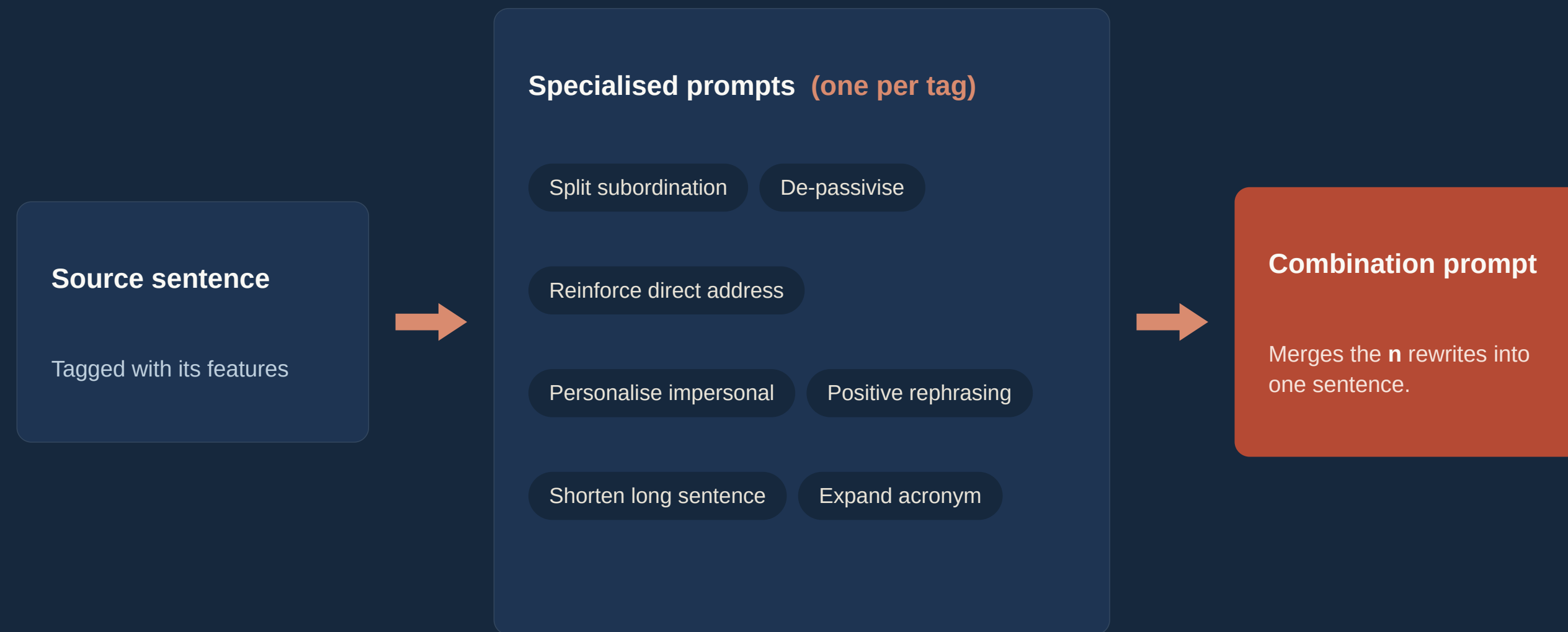
FA · HI · JA · KO · TH · ZH

Criteria unverified here. So we skip tagging and apply one all-guidelines prompt.

Taggers on the large spaCy models: one binary detector per feature, from POS, morphology and dependency cues.

PROMPTING STRATEGY

One prompt per phenomenon, then combine



One tag → use it directly. No tag → one all-guidelines prompt. Model: **Gemma 3 27B**, run locally on an NVIDIA L4 (24GB VRAM).

TWO SUBMITTED RUNS

Differing only in the combiner's input

Run 1 • **no-orig**

Combiner sees only the specialised outputs, not the source.

→ **More aggressive simplification**

Run 2 • **with-orig**

Combiner also sees the original sentence as an anchor.

→ **More content preserved, more conservative**

Everything else is identical across the two runs.

Choosing the model and the prompt

Model screening

Ten open-weight models on 50 dev sentences (L4 ruled out anything above 30B):

Gemma 3 (4B / 12B / 27B), GPT-OSS 20B, DeepSeek-R1 8B, Llama 3 & 3.1 8B, Llama 3.3 70B, Mistral 7B, Qwen3 8B.

GPT-OSS 20B led the surface metrics; Gemma 3 27B led readability. Both taken to the full evaluation.

Prompt design

Generic prompt → all-guidelines → tag-specific constraints.

Piling every guideline into one prompt improved SARI/BLEU but pushed FKGL from **11.2 to 14.3**. So we isolate one operation per call.

Model selection

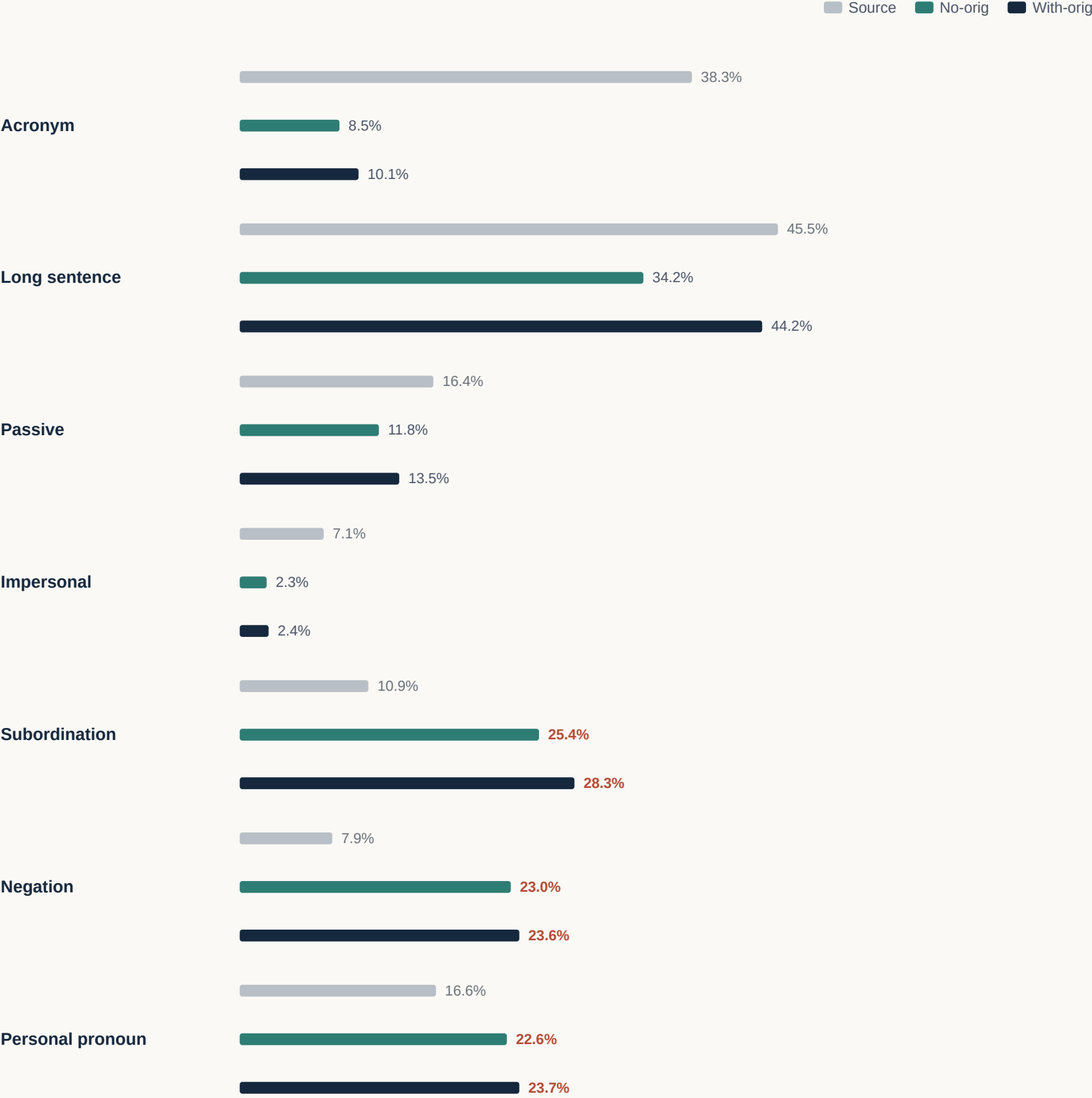
The full pipeline on the training set: two models × two combiner variants.

System	SARI	FKGL	BS-F1	CEFR Δ	Improved
GPT-OSS 20B · no-orig	38.68	11.82	0.316	+0.44	40.2%
GPT-OSS 20B · with-orig	37.02	12.45	0.292	+0.29	32.1%
Gemma 3 27B · no-orig (submitted)	36.71	11.53	0.285	+0.62	57.6%
Gemma 3 27B · with-orig (submitted)	36.02	12.11	0.264	+0.41	46.4%

GPT-OSS wins SARI and BERTScore (BS-F1), but Gemma 3 27B moves far more sentences to a simpler CEFR level. We optimise for reading level, so both submitted runs use Gemma; no-orig is the stronger of the two.

CEFR levels predicted with the classifier from the TSAR 2025 shared task.

How each feature's rate changes



OFFICIAL TEST SET · ENGLISH (175 PLS)

Test-set results

Run	SARI	BLEU	FKGL	Compr.	Splits	Leven.	Copies	Add.	Del.
Run 1 · no-orig	43.86	7.16	11.06	0.76	0.98	0.48	0.00	0.36	0.63
Run 2 · with-orig	44.20	7.80	12.71	0.88	0.98	0.51	0.00	0.40	0.55

Splits = sentence splits · Leven. = Levenshtein similarity · Copies = exact copies · Add./Del. = additions / deletions proportion.

LIMITATIONS

Two limitations

Subordination bug

Clauses were **detected** in English but the split prompt **never ran**. Splitting never fired; the combiner even added clauses, 11% → 25%.

Found only after submission, real impact unknown

Prompts interact

Negation tripled (8% → 23%): a by-product of the rewrites. Which transformations introduce it remains to investigate.

Isolating operations doesn't prevent the combiner from re-adding complexity.

TAKEAWAYS

Contribution

Taggers grounded in plain-language research → one prompt per feature → a combination step.

Outcome

SARI ~44 on the test set. During training >50% of sentences reach a simpler CEFR level.

Limitations

Negation and subordination rise: prompts interact, the combiner can sometimes re-add complexity.

Next

Penalise new complexity in the combiner; fix the split bug; extend cross-lingually.